

# altairsim — Quick Reference

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2026-08-21 · 157650b

## Getting out, and back in

| Key               | Does                                                                        |
|-------------------|-----------------------------------------------------------------------------|
| <code>^E</code>   | <b>ATTN</b> — stop the machine and take the keyboard back. Nothing is lost. |
| <code>RUN</code>  | Resume, at the exact instruction it stopped on.                             |
| <code>QUIT</code> | Leave. (There is no <code>EXIT</code> .)                                    |

## Editing the command line

`Tab` completes what you are typing — a command, then a board id, then its property names, then a property's values ( `SET mem0 fill=` then `Tab` ); a second `Tab` lists the choices. `Up` / `Down` walk the command history, saved per directory in `.altairsim_history`.

| Key                                                                                            | Does                                            |
|------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <code>Ctrl-A</code> / <code>Ctrl-E</code> (or <code>Home</code> / <code>End</code> )           | start / end of line                             |
| <code>Alt-B</code> / <code>Alt-F</code> (or <code>Ctrl-Left</code> / <code>Ctrl-Right</code> ) | back / forward one word                         |
| <code>Ctrl-W</code> / <code>Ctrl-K</code> / <code>Ctrl-U</code>                                | erase word behind / to end of line / whole line |
| <code>Backspace</code> / <code>Delete</code>                                                   | erase before / under the cursor                 |

## Command line

```
altairsim [options] [machine]
```

|                   |                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|
| machine           | a built-in name, or a config file (has a '/' or ends .toml).<br>Omitted: ./altairsim.toml if there is one, else `default`. |
| -m, --machine <n> | ALWAYS a built-in name -- never a file.                                                                                    |
| -f, --file <path> | ALWAYS a file -- never a built-in name.                                                                                    |
| -n, --none        | empty backplane: no boards, no memory, nothing.                                                                            |
| -l, --list        | list the built-in machines and exit.                                                                                       |
| -s, --script <f>  | run a command script, then exit with its status.                                                                           |
| -x, --exec <cmd>  | run one monitor command (repeatable), then exit.                                                                           |
| -i, --interactive | after --script/--exec, stay in the monitor.                                                                                |
| --mcp             | MCP server on stdio.                                                                                                       |
| -v, --version     | print the version and exit.                                                                                                |
| -h, --help        | print this help and exit.                                                                                                  |

## Monitor commands

Type the part before the bracket.

| Command      | Does                                                                                | Usage                                                                                                 |
|--------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| BO[ARDS]     | List, add, or remove boards on the backplane.                                       | BOARDS [LIST] ADD <type> <id> [k=v...] REMOVE <id>                                                    |
| B[REAK]      | Set a breakpoint on an address, memory/I/O access, or tape stop.                    | BREAK [<addr>   MEM R W <addr>   IO R W <port>   TAPE STOP] [IF <expr>   LOADS <expr>] [TRACE ON OFF] |
| COM[PARE]    | Compare a range of memory against another address.                                  | COMPARE <range> <addr>                                                                                |
| C[ONFIG]     | Load or save the whole machine as a TOML file.                                      | CONFIG LOAD <f.toml>   CONFIG SAVE <f.toml>                                                           |
| CONN[ECT]    | Attach a serial unit to an endpoint (console, socket, file, ...).                   | CONNECT <id>:<u> <endpoint>                                                                           |
| CONS[OLE]    | Show or set the host console's properties.                                          | CONSOLE [<k>=<v>...]                                                                                  |
| DE[POSIT]    | Write bytes into memory at an address.                                              | DEPOSIT <addr> <bytes...>                                                                             |
| DI[SASM]     | Disassemble memory into instructions.                                               | DISASM [<addr> <range>] [n] [CPU=8080]                                                                |
| DISC[ONNECT] | Unplug the endpoint from a serial unit.                                             | DISCONNECT <id>:<u>                                                                                   |
| D[UMP]       | Show memory as hex and ASCII.                                                       | DUMP [<addr> <range>] [WIDTH=16]                                                                      |
| E[EDIT]      | Enter bytes into memory interactively from an address.                              | EDIT <addr> [ROM]                                                                                     |
| EX[AMINE]    | Point the front panel at an address (and show that byte).                           | EXAMINE [<addr>]                                                                                      |
| F[ILL]       | Fill a range of memory with a byte.                                                 | FILL <range> <byte>                                                                                   |
| HE[LP]       | Show help for a command.                                                            | HELP [<command>]                                                                                      |
| H[ISTORY]    | Replay the recent instruction (or bus-cycle) history.                               | HISTORY [BUS CPU] [n]                                                                                 |
| I[N]         | Read a byte from an I/O port.                                                       | IN <port>                                                                                             |
| L[OAD]       | Load a file into memory (binary or Intel hex).                                      | LOAD <file> [AT <addr>] [FORMAT=BIN HEX] [ROM]                                                        |
| M[OUNT]      | Put a disk or tape image into a drive; a .imd is converted to a raw .dsk beside it. | MOUNT <id>[:<u>] <file> [WP] [CREATE] [extract[=<base>]] [k=v...]                                     |
| MOV[E]       | Copy a range of memory to another address.                                          | MOVE <range> <dest> [ROM]                                                                             |

|           |                                                                       |                                                          |
|-----------|-----------------------------------------------------------------------|----------------------------------------------------------|
| N[EXT]    | Step one instruction, running any CALL/RST to completion.             | NEXT                                                     |
| NO[BREAK] | Remove a breakpoint, or all of them.                                  | NOBREAK [id]                                             |
| O[UT]     | Write a byte to an I/O port.                                          | OUT <port> <byte>                                        |
| P[OWER]   | Power-cycle the machine -- the only thing that clears RAM.            | POWER                                                    |
| Q[UIT]    | Leave the simulator.                                                  | QUIT                                                     |
| REGI[ON]  | Add a RAM or ROM region to a memory board.                            | REGION ADD <id> type=ram rom at=<addr><br>[size= mount=] |
| RE[GS]    | Show the CPU registers (SET REG changes one).                         | REGS   SET REG <r>=<v>                                   |
| RES[ET]   | Reset the machine, keeping RAM (RESET CPU resets just the processor). | RESET [CPU]                                              |
| REST[ORE] | Load machine state back from a snapshot.                              | RESTORE <file>                                           |
| R[UN]     | Start or resume the machine, optionally at an address.                | RUN [addr]                                               |
| SA[VE]    | Write a range of memory out to a file.                                | SAVE <file> <range><br>[FORMAT=BIN HEX OCTAL PRN]        |
| SEA[RCH]  | Find bytes or a string in a                                           |                                                          |

|                         |                                                                             |                                                                                                                                                                              |
|-------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | range of memory.                                                            | <code>SEARCH &lt;range&gt; &lt;bytes...&gt; "str"</code>                                                                                                                     |
| <code>SE[T]</code>      | Change a property of a board, the console, display, a register, or the bus. | <code>SET &lt;id&gt;[:&lt;u&gt;] CONSOLE DISPLAY REG BUS &lt;k&gt;=&lt;v&gt;</code>                                                                                          |
| <code>SH[OW]</code>     | Display the state of a board, the bus, or the machine.                      | <code>SHOW &lt;id&gt; BOARDS BOARD &lt;type&gt; [UNITS] MACHINES MACHINE [&lt;name&gt;] BUS [MAP IO IRQ CONTENTION] ROMS MOUNTS PATHS CONSOLE DISPLAY SYMBOLS VERSION</code> |
| <code>SN[APSHOT]</code> | Save the whole machine state to a file.                                     | <code>SNAPSHOT &lt;file&gt;</code>                                                                                                                                           |
| <code>S[TEP]</code>     | Run one instruction (or n), showing the registers after each.               | <code>STEP [n]</code>                                                                                                                                                        |
| <code>SY[MBOLS]</code>  | Load or clear a symbol table for disassembly.                               | <code>SYMBOLS LOAD &lt;file&gt; [REPLACE]   SYMBOLS CLEAR</code>                                                                                                             |
| <code>T[RACE]</code>    | Log every bus cycle while the machine runs.                                 | <code>TRACE ON OFF [file] [MASK=IN,OUT,IRQ,DMA,CONTENTION]</code>                                                                                                            |
| <code>TY[PE]</code>     | Feed text to the guest as if typed at its keyboard.                         | <code>TYPE "text"</code>                                                                                                                                                     |
| <code>U[NMOUNT]</code>  | Take a disk or tape out of a drive.                                         | <code>UNMOUNT &lt;id&gt;:&lt;u&gt;</code>                                                                                                                                    |
| <code>W[H0]</code>      | Say which board answers an address or I/O port.                             | <code>WHO &lt;addr&gt;   WHO IO &lt;port&gt;</code>                                                                                                                          |

## Boards

### CPU

| Type | What it is                                                                                                                                        |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 6800 | Altair 680b CPU board: a Motorola 6800 at 500 KHz. Decodes nothing -- it drives the bus. The 88-CPU's twin, one core down, with memory-mapped I/O |
| 8080 | MITS 88-CPU: an 8080A at 2 MHz. Decodes nothing -- it drives the bus                                                                              |
| 8085 | Generic 8085 CPU board. Decodes nothing -- it drives the bus. The 88-CPU's twin, with an 8085 core (RIM/SIM + TRAP/RST 5.5/6.5/7.5)               |
| z80  | Generic Z80 CPU board. Decodes nothing -- it drives the bus. The 88-CPU's twin, with a Z80 core                                                   |

### Memory

| Type     | What it is                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bankmem  | S-100 bank-switched RAM. One card, four decoders (card=vector cromemco64kz northstar expandoram2): a write-only select port swaps which RAM plane(s) drive the bus. Each card owns its own decode -- one-hot select (Vector 40), 8-bit bank mask (Cromemco 40), on/off+one-hot toggle (North Star C0), or PROM page-select (ExpandoRAM II FF, approximated)                                                                   |
| memory   | RAM/ROM board: a list of regions and PHANTOM* -- plain, unbanked memory (bank switching is its own board, bankmem )                                                                                                                                                                                                                                                                                                           |
| v2z80rom | S100Computers V2 Z80 CPU board -- its onboard paged monitor EEPROM (the Z80 itself is board 'z80cpu'). An 8K 28C64 at F000-FFFF holding two 4K pages, builtin:master0 (low) / master1 (high), selected by OUT D3H bit1 (bit0=1 inactivates the EEPROM so RAM shows through). Shadows RAM in its window while enabled. Cold-start the MASTER monitor with startup=["RUN F000"]; the 'I' command boots CP/M 3 off a dualsd card |

## Disk

| Type        | What it is                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16fdc       | Cromemco 16FDC: WD FD1793 soft-sector floppy (single + double density), up to 4 drives. Disk registers at 30-34, a TMS 5501 console UART at 00-09 (unit 'tty'), and a 4K RDOS 2.52 boot PROM at C000 (OUT 40H banks it out, RESET restores it). Boots CDOS                                                                                                                                                                                                                                                      |
| 64fdc       | Cromemco 64FDC: the 16FDC's 1983 successor -- same FD1793 + TMS 5501, carrying an 8K RDOS 3.12 boot PROM at C000-DFFF (OUT 40H banks it out, RESET restores it). Boots CDOS                                                                                                                                                                                                                                                                                                                                     |
| dcdd        | MIT 88-DCDD: 8" hard-sector floppy, up to 16 drives. Three ports at BASE+0..2. INVERTED status bits                                                                                                                                                                                                                                                                                                                                                                                                             |
| dualide     | S100Computers IDE-AB (CF): the IDE/CompactFlash half of the IDE+ESP32 combination board -- two CF sockets (drives 0/1 = A:/B:) for CP/M 3. Five 8255 ports at BASE+0..4 (default 30): A/B data, C control lines, mode config, drive select. Programmed-I/O ATA register engine (LBA read/write, 512-byte sectors). No boot PROM -- the CPU board's monitor boots CP/M from the CF. Mounts the SAME card image as dualsd (a .img with a .geo geometry sidecar); pair with dualsd for the full A:/B:/C:/D: system |
| dualsd      | S100Computers Dual SD: two microSD sockets (drives 0/1) presented as raw 512-byte-sector CF/SD cards, for CP/M 3. Two ports at BASE+0..1 (default 80): status/command + data. Programmed-I/O command/handshake engine (33H-lead + 8 commands). No boot PROM -- the CPU board's monitor loads CP/M from track 0. Mount a card image (a .img with a .geo geometry sidecar)                                                                                                                                        |
| hdsk        | MIT 88-HDSK Datakeeper: Pertec hard disk, 256-byte sectors from a linear .DSK. Eight ports at BASE+0..7 (default A0). Command/handshake protocol, four page buffers                                                                                                                                                                                                                                                                                                                                             |
| icom        | iCOM FD3712/FD3812 8" floppy: a programmed-I/O command/handshake controller on the S-100 Interface board. Two ports at BASE+0..1 (default C0) plus a boot PROM and 6810 scratch RAM in high memory (rom=builtin:icom-fd3712-cpm   icom-fd3712-fdos   icom-fd3812-cpm). Boots CP/M 2.2 (single and double density) and FDOS. Up to 4 drives                                                                                                                                                                      |
| mds         | MIT 88-MDS: 5.25" minidisk, 4 drives. Same three ports as the dcdd -- but 300 RPM, 64 us/byte, and a motor that stops after 6.4 s                                                                                                                                                                                                                                                                                                                                                                               |
| tarbell     | Tarbell #1011: single-density WD FD1771 floppy, up to 4 drives. Eight ports at BASE+0..7 (default F8). Carries a 32-byte boot PROM that shadows 0000 over PHANTOM* -- boots CP/M automatically at reset (bootstrap=on)                                                                                                                                                                                                                                                                                          |
| tarbelldd   | Tarbell #2022: double-density WD FD1791 floppy (mixed-density media, SD track 0), up to 4 drives. The single-density card's twin with a bitmap OUT-FC latch and a port-FD DMA/ext-addr register. Same 32-byte boot PROM                                                                                                                                                                                                                                                                                         |
| versafloppy | SD Systems VersaFloppy I/II: WD FD177x soft-sector floppy, up to 4 drives. Eight ports at BASE+0..7 (default 60). variant=vfi (FD1771, single density)   vfii (FD1791, single+double). Boots SDOS with the SBC-200 + DDBIOS                                                                                                                                                                                                                                                                                     |

## Serial

| Type    | What it is                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2sio    | MITs 88-2SIO: two 6850 ACIAs, units 'a' and 'b'. Four ports at BASE+0..3                                                                                                                                                                                                                                                                                                                                                                                                               |
| 680io   | Altair 680b onboard I/O: a 6850 ACIA console ('tty') at F000/F001 and the config-strap read port at F002. Memory-mapped                                                                                                                                                                                                                                                                                                                                                                |
| pmmi    | PMMI MM-103: Bell 103 modem on an S-100 card, unit 'line'. Four ports at BASE+0..3 (default C0), read/write different registers. Transmit/receive over a ByteStream; CONNECT it to in:/out: files. No dialer; modem status is a fixed stub                                                                                                                                                                                                                                             |
| propio  | S100Computers Console IO Board (Parallax-Propeller console), unit 'serial'. A usio subtype preset to the board's documented convention: status/data at 00/01, RX-ready = status bit 1, TX-ready = status bit 2, both active high. Every strap (status_port/data_port/rdr_bit/tdre_bit/polarity) is still overridable -- the real board is jumpered. Polled, no interrupts. CONNECT it to a file, socket, serial port, in:/out:                                                         |
| sbc     | SD Systems SBC-100/200: Z80 single-board computer. One 8-port block (78-7F): Intel 8251 console (unit 'tty', data 7C / status 7D, Rx/D->/DSR auto-baud for MSMONR21), Z80-CTC (78-7B) whose ch1 raises a mode-2 keyboard interrupt (vector 0x82) off the 8251 RxRDY, and a parallel port (7E/7F) whose OUT 7F bit 1 switches the onboard PROM out. Optional onboard boot PROM via [[board.socket]] (at+mount). variant=sbc100 sbc200                                                   |
| sio     | MITs 88-SIO: one COM2502 UART, unit 'tty'. Two ports at BASE+0..1. INVERTED status bits                                                                                                                                                                                                                                                                                                                                                                                                |
| turnkey | MITs 8800b Turnkey Module: phantom boot PROM (FC00-FFFF), integrated 6850 SIO (unit 'tty', default 0x10), sense switches at FF, and the Auto-Start JMP jam. Sockets via [[board.socket]]                                                                                                                                                                                                                                                                                               |
| usio    | Universal Serial board: a UART-agnostic serial card, unit 'serial'. Two ports you strap: a status/control port (status_port -- read synthesizes RDR/TDRE at bit positions you pick, write is discarded) and a data port (data_port). Built-in profiles preset the straps: profile=tuart (Cromemco TU-ART)   imesai-sio2   compupro-if2 (CompuPro Interfacer II)   compupro-ss1 (CompuPro System Support 1). Polled, no interrupts. CONNECT it to a file, socket, serial port, in:/out: |

## Tape

| Type     | What it is                                                                                                                                                                                                                                                                                                                                                   |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 680kcacr | Altair 680b KCACR audio-cassette interface: a 1602-family UART recording Kansas City Standard FSK, memory-mapped at F010 (status/control) and F011 (data), active-LOW. Adds software motor control (control D7=on, D6=off) and interrupt-driven transfer (D0/D1 enables pull the 6800 IRQ). Reuses the 88-ACR tape machinery -- MOUNT a tape, WIND/REWIND it |
| acr      | MITs 88-ACR: cassette. An 88-SIO B + an FSK modem, unit 'tape'. Brings the WIND/REWIND/EXTRACT verbs and a tape counter                                                                                                                                                                                                                                      |
| uio      | MITs 88-UIO: serial + cassette on one board. A 6850 (unit 'serial', default 0x10) and an 88-ACR cassette section (unit 'tape', default 0x06) with motor control and a SW-1 MITs/Kansas-City modulation switch. Defaults reproduce the standard 0x10 + 0x06 layout                                                                                            |

## Parallel and printer

| Type   | What it is                                                                                                                                                                                                                                                                                                                         |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4pio   | MIT 88-4PIO: up to four 6820 PIAs, sections ja/jb.. per port. 16 ports from BASE (default 20). Software-set direction; CONNECT each section                                                                                                                                                                                        |
| 680uio | Altair 680b Universal I/O: a second 6850 ACIA serial port ('serial') and a 6820 PIA parallel port (sections 'p1a/p1b', 'p2a/p2b' with pias=2) in an S9-relocatable window (default base F000: serial F006/F007, PIA F008-F00F), plus fixed switch inputs at F003 and a non-latched output at F010-F013. Memory-mapped, active-high |
| c700   | MIT 88-C700: Centronics line-printer controller, unit 'prn'. Two ports at BASE+0..1 (default 02). Output-only; CONNECT it to a file, a socket, or a real printer queue                                                                                                                                                             |
| d7a    | Cromemco D+7A: analog + parallel I/O. Eight ports from BASE (default 18): one parallel port + seven two's-complement A/D-in/D/A-out channels. Reads 1-2 JS-1 joysticks from the host                                                                                                                                               |
| lpc    | MIT 88-LPC: 88-LP line-printer controller, unit 'prn'. Two ports at BASE+0..1 (default 02). Line-buffered: 6-bit codes + PRINT/LINE FEED/CLEAR. CONNECT it to a file, a socket, or a real printer queue                                                                                                                            |
| pio    | MIT 88-PIO: 8-bit parallel port, units 'out'/'in'. Two ports at BASE+0..1 (default 04). CONNECT a printer, a keyboard, a socket                                                                                                                                                                                                    |

## Video

| Type    | What it is                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dazzler | Cromemco Dazzler: color graphics from a framebuffer in main RAM. Two ports at BASE+0..1 (default 0E): control/status and format. 32x32 to 128x128, 16 colors/greys. Needs a Display                                                                                                                                                                                                                             |
| vdb8024 | SD Systems VDB-8024: an 80x24 video terminal on one board -- the video console for an SBC-100/200 (the alternative to the 8251). Two I/O ports at BASE+0..1 (default 00): status/keyboard/display. Unit 'keyboard' (CONNECT). Optional keyboard-strobe interrupt strap (interrupt=vi0..vi7) for the SBC-200's CTC to vector - what the SD video CBIOS needs; polled by default. Boots sdmonv21. Needs a Display |
| vdm1    | Processor Technology VDM-1: memory-mapped 16x64 video, screen RAM at BASE (default CC00), scroll/status port (default CC). Needs a Display                                                                                                                                                                                                                                                                      |

## Systems

| Type | What it is                                                                                                                                                                                                                        |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sol  | Processor Technology Sol-PC I/O: serial, keyboard, parallel, CUTS tape as one board. Seven ports F8..FE. Units serial/printer/keyboard (CONNECT) and tape1/tape2 (MOUNT). Brings the WIND/REWIND/EXTRACT verbs and a tape counter |

## Other

| Type       | What it is                                                                                                                              |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| fp         | Altair front panel: the SENSE switches a guest reads at IN 0FFH -- a configured byte (SET fp0 sense= or TOML), not toggled here. No OUT |
| hostbridge | Host Bridge: guest <-> host file transfer, sandboxed. OUR OWN BOARD, not a period one. Two ports at BASE+0..1. R.COM/W.COM/HDIR.COM     |
| virtc      | MIT 88-VI/RTC: vectored interrupts (VI0-VI7 -> RST n) and a real-time clock. One port at FE                                             |

## Machines

| Machine         | What it is                                                                                                           |
|-----------------|----------------------------------------------------------------------------------------------------------------------|
| 8085            | A minimal 8085 machine: an 8085 CPU, 64K of RAM, and a 2SIO console.                                                 |
| acuter          | ACUTER at F000 -- CUTER on a plain Altair, with a terminal instead of a VDM.                                         |
| altair680       | The Altair 680b -- MITS's second machine, and a different animal from the 8800.                                      |
| altmon          | An Altair with a monitor in ROM and a terminal on it.                                                                |
| amon            | AMON 3.1 in a 4K EPROM at F000 -- Martin Eberhard's full-featured Altair monitor.                                    |
| bankmem         | A bank-switched RAM machine: a Z80, a console, and a Vector Graphic 64K bankmem.                                     |
| basic4k         | The machine Altair 4K BASIC was sold to run on: an 88-SIO Teletype, a cassette in the ACR.                           |
| basic8k         | The machine Altair 8K BASIC was sold to run on: an 88-2SIO terminal, a cassette in the ACR.                          |
| cdbl            | The default machine with the Combo Disk Boot Loader in the PROM socket.                                              |
| cuter           | CUTER 1.3 driving a Processor Technology VDM-1 -- the real Sol/CUTS monitor.                                         |
| dazzler         | A Cromemco Dazzler in an Altair -- the bench for the S-100's first color graphics card.                              |
| default         | The machine you get when you name none: 56K, and the DBL boot PROM at FF00.                                          |
| dualide         | S100Computers "IDE-AB CF" machine -- a V2 Z80 CPU board booting CP/M 3 off a CompactFlash card.                      |
| dualidesd       | S100Computers "IDE-AB CF+ESP32" machine -- both boards, CP/M 3 on CF drives A:/B: and SD drives C:/D:.               |
| dualsd          | S100Computers "Dual SD" machine -- a V2 Z80 CPU board booting CP/M 3 off a microSD card.                             |
| icom            | iCOM FD3712 8" floppy machine -- boots CP/M 2.2 off a single-density iCOM disk.                                      |
| lineprinter-lpc | The default machine with an 88-LPC line printer at port 02; CONNECT <code>lpt0:prn</code> to a file or the console.  |
| lineprinter     | The default machine with an 88-C700 line printer at port 02; CONNECT <code>lpt0:prn</code> to a file or the console. |
| minidisk        | The Altair Minidisk: an 88-MDS at 08 and the MDBL boot PROM. You supply the 5.25" disk.                              |
| original        | The Altair as it actually left Albuquerque.                                                                          |
| parallel        | The default machine with two MITS parallel boards: an 88-PIO and an 88-4PIO.                                         |
| ps2             | The machine MITS Programming System II ran on: basic8k's cards, but not basic8k's bootstrap.                         |
| ps2int          | MITS Programming System II, WITH INTERRUPTS. ps2 with A9 down and an 88-VI/RTC in it.                                |
| rombasic        | MITS Extended ROM BASIC 16K -- Extended BASIC that runs directly out of ROM.                                         |
| sbc200          | SD Systems SBC-200 -- a 4 MHz Z80 single-board computer running the MSMONR21 monitor.                                |

|           |                                                                                           |
|-----------|-------------------------------------------------------------------------------------------|
| sbc200v   | The SD Systems SBC-200 with a VDB-8024 VIDEO console: SDMONV21 on an 80x24 screen.        |
| sol20     | The Processor Technology Sol-20 -- an integrated 8080 machine, running SOLOS.             |
| tarbell   | Tarbell #1011 single-density floppy machine -- boots CP/M 2.2 the moment a disk is in it. |
| tarbelldd | Tarbell #2022 double-density floppy machine -- boots CP/M 2.2 the moment a disk is in it. |
| turnkey   | The MITS 8800bt -- an Altair with a Turnkey Module where the front panel used to be.      |
| vdm1      | A Processor Technology VDM-1 in an Altair, and a demo that draws on it.                   |
| z80       | A minimal Z80 machine: a z80 CPU, 64K of RAM, and a 2SIO console.                         |

## A machine file, in one look

```
[machine]
name      = "mine"
base      = "default"      # start from a machine, and say what is DIFFERENT
startup   = ["RUN FF00"]    # the operator's own keystrokes. There is no BOOT verb.

[[board]]                # type + a NEW id      -> ADD the card
type       = "2sio"        # type + an id from the base -> REPLACE it outright
id         = "sio0"        # NO type + an id      -> MODIFY the one already there
port       = 10            # remove = true       -> PULL THE CARD OUT

[board.unit.a]           # a unit's own settings
connect    = "console"

[[board.region]]         # a list the card owns (memory)
type       = "ram"
at         = 0000          # hex: it is an address
size       = "56K"         # decimal: it is a size

[[board.drive]]          # a list the card owns (disk controllers)
unit       = 0
mount      = "cpm.dsk"     # relative to THIS FILE

[console]                # the HOST's terminal -- not a board
strip7out  = true
base       = octal         # read/print the wire class in split octal (MITS style)
```

**Paths:** a path *inside* a machine file is relative to **that file**. A path you *type* is relative to **your shell**.

## Endpoints — `CONNECT <id>:<unit> <endpoint>`

| Endpoint                           | Is                                                                                                                |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <code>console</code>               | the host terminal. Exactly one unit may hold it.                                                                  |
| <code>null</code>                  | nowhere. Writes vanish, reads never come.                                                                         |
| <code>loopback</code>              | itself — what you write comes back.                                                                               |
| <code>scripted</code>              | a caller in place of a human — what MCP and the tests type into.                                                  |
| <code>socket:PORT</code>           | <b>listens</b> — this is telnet-in.                                                                               |
| <code>socket:HOST:PORT</code>      | <b>calls out</b> .                                                                                                |
| <code>serial:DEVICE</code>         | a real serial port on this host.                                                                                  |
| <code>in:PATH</code>               | a host file as a reader (paper tape). <code>?cps=N</code> paces it.                                               |
| <code>out:PATH</code>              | a host file as a punch — 8-bit clean, never truncating.                                                           |
| <code>terminal</code>              | a window the simulator draws itself (SDL builds). <code>?emulation=vt100 adm3a vt52 h19, ?size=COLSxROWS</code> . |
| <code>printer:QUEUE</code>         | a real print queue on this host.                                                                                  |
| <code>&lt;endpoint&gt; FILE</code> | a tap: append <code> FILE</code> to any endpoint to also log the line.                                            |